Oregon Department of Environmental Quality

OAR 340-253-8010
Tables

Table 1 Oregon Clean Fuel Standard for Gasoline and Gasoline Substitutes		
Calendar Year	**Oregon Clean Fuel Standard (gCO2e per MJ)**	**Percent Reduction**
2015	None (Gasoline Baseline is 98.62 for 2016-2017, 98.64 for 2018, 98.06 for 2019-2025, and 96.50 for 2026 and beyond)	
2016*	98.37	0.25 percent
2017	98.13	0.50 percent
2018	97.66	1.00 percent
2019	96.59	1.50 percent
2020	95.61	2.50 percent
2021	94.63	3.50 percent
2022	93.15	5.00 percent
2023	91.68	6.50 percent
2024	90.21	8.00 percent
2025	88.25	10.00 percent
2026	84.92	12.00 percent
2027	82.99	14.00 percent
2028	81.06	16.00 percent
2029	79.13	18.00 percent
2030	77.20	20.00 percent
2031	73.92	23.40 percent
2032	70.64	26.80 percent
2033	67.36	30.20 percent

Table 1 Oregon Clean Fuel Standard for Gasoline and Gasoline Substitutes		
2034	64.08	33.60 percent
2035 and beyond	60.80	37.00 percent
*Initial compliance period is a two-year period for 2016 and 2017.		

Table 2
Oregon Clean Fuel Standard for Diesel Fuel and Diesel Substitutes

Calendar Year	Oregon Clean Fuel Standard (gCO2e per MJ)	Percent Reduction
2015	None (Diesel Baseline is 99.64 for 2016-2017, 99.61 for 2018, 98.74 for 2019-2025, and 102.73 for 2026 and beyond)	
2016*	99.39	0.25 percent
2017	99.14	0.50 percent
2018	98.61	1.00 percent
2019	97.26	1.50 percent
2020	96.27	2.50 percent
2021	95.29	3.50 percent
2022	93.81	5.00 percent
2023	92.32	6.50 percent
2024	90.84	8.00 percent
2025	88.87	10.00 percent
2026	90.40	12.00 percent
2027	88.34	14.00 percent
2028	86.29	16.00 percent
2029	84.24	18.00 percent
2030	82.18	20.00 percent
2031	78.69	23.40 percent
2032	75.20	26.80 percent
2033	71.70	30.20 percent
2034	68.21	33.60 percent
2035 and beyond	64.72	37.00 percent

*Initial compliance period is a two-year period for 2016 and 2017.

Table 3
Oregon Clean Fuel Standard for Alternative Jet Fuel

Calendar Year	Oregon Clean Fuel Standard (gCO2e per MJ)
2015	None (Diesel Baseline is 99.64 for 2016-2017, 99.61 for 2018, and 98.74 for 2019 and beyond. The fossil jet baseline is 88.21.)
2019	90.80
2020	90.80
2021	90.80
2022	90.80
2023	90.80
2024	90.80
2025	88.87
2026	88.21
2027	88.21
2028	86.29
2029	84.24
2030	82.18
2031	78.69
2032	75.20
2033	71.70
2034	68.21
2035 and beyond	64.72

Table 4
Oregon Carbon Intensity Lookup Table

Fuel	Pathway Identifier	Pathway Description	Carbon Intensity Values (gCO2e/MJ)
			Total Lifecycle Emissions
Gasoline	ORGAS001	Clear gasoline - based on a weighted average of gasoline supplied to Oregon	100.14 for 2025, 98.12 for 2026 and beyond
	ORGAS002	Imported blended gasoline (E10) – 90% clear gasoline & 10% corn ethanol based on Midwest average. Cannot be used to report exports except when the specific gallon was also imported under this fuel pathway code.	98.06 for 2025, 96.50 for 2026 and beyond
Diesel	ORULSD001	Clear diesel, based on a weighted average of diesel fuel supplied to Oregon	100.74 for 2025, 104.92 for 2026 and beyond
	ORULSD002	Imported blended diesel (B5) – 95% clear diesel & 5% soybean biodiesel. Cannot be used to report exports except when the specific gallon was also imported under this fuel pathway code.	98.74 for 2025, 102.73 for 2026 and beyond
	ORULSD003	Imported blended diesel (B20) – 80% clear diesel & 20% soybean biodiesel. Cannot be used to report exports except when the specific gallon was also imported under this fuel pathway code.	92.68 for 2025, 96.06 for 2026 and beyond
Compressed Natural Gas	ORCNG001	North American NG delivered via pipeline; compressed in OR	79.98 for 2025, 81.89 for 2026 and beyond
Liquefied Petroleum Gas	ORLPG001	Liquefied petroleum gas	80.88 for 2025, 81.69 for 2026 and beyond
Electricity	ORELEC100	Solar power, produced at or directly connected to the site of the charging station in Oregon, subject to OAR 340-253-0470 (3).	0
	ORELEC101	Wind power, produced at or directly connected to the site of the charging station in Oregon, subject to OAR 340-253-0470 (3).	0
	ORELEC200	Renewable power deemed to have a carbon intensity of zero under OAR 340-253-0470 and meeting the provisions of (5).	0

Table 5
Summary Checklist of Quarterly and Annual Compliance Reporting Requirements

Parameters to Report	Gasoline & Diesel Fuel	Ethanol, Biodiesel & Renewable Diesel	CNG, LNG & LPG	Electricity	Hydrogen & Hydrogen Blends
Company or organization name	x	x	x	x	x
Reporting period	x	x	x	x	x
Fuel pathway code	x	x	x	x	x
Transaction type	x	x	x	x	x
Transaction date	x	x	x	x	x
Business Partner	x	x	x	x	x
Production Company ID and Facility ID	n/a	x	n/a	n/a	x
Physical transport mode code	x	x	x	x	x
Aggregation	x	x	x	x	x
Application / EER	x	x	x	x	x
Amount of each fuel used as gasoline replacement	x	x	x	x	x
Amount of each fuel used as diesel fuel replacement	x	x	x	x	x
*Credits/deficits generated per quarter (MT)	x	x	x	x	x
For Annual Compliance Reporting (in addition to the items above)					
*Credits and Deficits generated per year (MT)	x	x	x	x	x
*Credits/deficits carried over from the	x	x	x	x	x

Table 5
Summary Checklist of Quarterly and Annual Compliance Reporting Requirements

Parameters to Report	Gasoline & Diesel Fuel	Ethanol, Biodiesel & Renewable Diesel	CNG, LNG & LPG	Electricity	Hydrogen & Hydrogen Blends
previous year (MT), if any					
*Credits acquired from another party (MT), if any	x	x	x	x	x
*Credits sold to another party (MT), if any	x	x	x	x	x
*Credits retired within the CFP (MT) to meet a compliance obligation, if any	x	x	x	x	x

Table 6
Oregon Energy Densities of Fuels

Fuel (unit)	MJ/unit
Gasoline (gallon)	122.48 (MJ/gallon)
Diesel fuel (gallon)	134.48 (MJ/gallon)
Compressed natural gas (therm)	105.5 (MJ/therms)
Electricity (kilowatt hour)	3.60 (MJ/kilowatt hour)
Denatured ethanol (gallon)	81.51 (MJ/gallon)
Clear biodiesel (gallon)	126.13 (MJ/gallon)
Liquefied natural gas (gallon)	78.83 (MJ/gallon)
Hydrogen (kilogram)	120.00 (MJ/kilogram)
Liquefied petroleum gas (gallon)	89.63 (MJ/gallon)
Renewable hydrocarbon diesel (gallon)	129.65 (MJ/gallon)
Undenatured anhydrous ethanol (gallon)	80.53 (MJ/gallon)
Alternative Jet Fuel (gallon)	126.37 (MJ/gallon)
Renewable naphtha (gallon)	117.66 (MJ/gallon)

Table 7
Oregon Energy Economy Ratio Values for Fuels

Light/Medium Duty Applications (Fuels used as gasoline replacements)		Heavy-Duty/Off-Road Applications (Fuels used as diesel replacements)		Aviation Applications (Fuels used as jet fuel replacements)	
Fuel/Vehicle Combination	**EER Value Relative to Gasoline**	**Fuel/Vehicle Combination**	**EER Value Relative to Diesel**	**Fuel/Vehicle Combination**	**EER Value Relative to conventional jet**
Gasoline (including E10) or any other gasoline-ethanol blend	1	Diesel fuel (including B5) or any other blend of diesel and biodiesel or renewable hydrocarbon diesel	1	Alternative Jet Fuel	1
CNG Internal Combustion Engine Vehicle (ICEV)	1	CNG, LNG, or LPG (Spark-Ignition Engines)	0.9	- - -	
Electricity/Battery Electric Vehicle or Plug-In Hybrid Electric Vehicle	3.4	CNG, LNG, or LPG (Compression-Ignition Engines)	1		
Electricity/On-Road Electric Motorcycle	4.4	Electricity/Battery Electric Vehicle or Plug-In Hybrid Electric Vehicle	5		
Propane/Propane Forklift	0.9	Electricity/Battery Electric or Plug-in Hybrid Transit Bus	5		
Hydrogen/Fuel Cell Vehicle	2.5	Electricity/Fixed Guideway Light Rail	3.3		
Electricity/Ground Support Equipment	3.2	Electricity/Fixed Guideway Streetcar	2.1		
- - -		Electricity/Fixed Guideway Aerial Tram	2.6		
		Electricity/Electric Forklift	3.8		
		Electricity/Electric TRU (eTRU)	3.4		

Table 7
Oregon Energy Economy Ratio Values for Fuels

Light/Medium Duty Applications (Fuels used as gasoline replacements)		Heavy-Duty/Off-Road Applications (Fuels used as diesel replacements)		Aviation Applications (Fuels used as jet fuel replacements)	
Fuel/Vehicle Combination	**EER Value Relative to Gasoline**	**Fuel/Vehicle Combination**	**EER Value Relative to Diesel**	**Fuel/Vehicle Combination**	**EER Value Relative to conventional jet**
		Hydrogen/Fuel Cell Vehicle	1.9		
		Hydrogen/Fuel Cell Forklift	2.1		
		Electricity/Cargo Handling Equipment	2.7		
		Electricity/Ocean Going Vessels	2.6		

Table 8
Oregon Substitute Fuel Pathway Codes

Fuel	Fuel Pathway code	CI (gCO2e/MJ)
Substitute CI for Ethanol. This pathway may only be used to report transactions that are sales or purchases without obligation, exports, loss of inventory, not for transportation use, and exempt fuel use.	ETH0116	40
Substitute CI for Biodiesel. This pathway may only be used to report transactions that are sales or purchases without obligation, exports, loss of inventory, not for transportation use, and exempt fuel use.	BIOD0116	15
Substitute CI for Renewable Diesel. This pathway may only be used to report transactions that are sales or purchases without obligation, exports, loss of inventory, not for transportation use, and exempt fuel use.	RNWD0116	15
Substitute CI for E10 Gasoline. This pathway may only be used to report transactions that are sales or purchases without obligation, exports, loss of inventory, not for transportation use, and exempt fuel use.	ORGAS0116	For 2019: 96.59 For 2020-2025: 96.00 For 2026 and beyond: 94.12
Substitute CI for B5 Diesel. This pathway may only be used to report transactions that are sales or purchases without obligation, exports, loss of inventory, not for transportation use, and exempt fuel use.	ORULSD01165	For 2019: 97.26 For 2020-2025: 96.71 For 2026 and beyond: 101.64
Substitute CI for B20 Diesel. This pathway may only be used to report transactions that are sales or purchases without obligation, exports, loss of inventory, not for transportation use, and exempt fuel use.	ORULSD011620	For 2020-2025: 84.45 For 2026 and beyond: 87.84

Table 9
Oregon Temporary Fuel Pathway Codes for Fuels with Indeterminate CIs

Fuel	Feedstock	Process Energy	FPC	CI (gCO_2e/MJ)
Ethanol	Corn	Grid electricity, natural gas, and/or renewables	ORETH100T	77.8
	Sorghum	Grid electricity, natural gas, and/or renewables	ORETH101T	95
	Sugarcane and Molasses	Bagasse and straw only, no grid electricity	ORETH102T	60
	Any starch or sugar feedstock	Any	ORETH103T	For 2025: 100.14, for 2026 and beyond: 98.12
	Corn Stover, Wheat Straw, or Sugarcane Straw	As specified in OR-Greet 2.0	ORETH104T	50
Biodiesel	Any feedstock derived from animal fats, corn oil, or a waste stream	Grid electricity, natural gas, and/or renewables	ORBIOD200T	50
	Any feedstock derived from plant oils except for Palm-derived oils	Grid electricity, natural gas, and/or renewables	ORBIOD201T	70
	Distiller's Corn Oil	Grid electricity, natural gas, and/or renewables	ORBIOD203T	60
	Any feedstock	Any	ORBIOD202T	For 2025: 100.74 For 2026 and beyond: 104.92
Renewable Diesel	Any feedstock derived from animal fats, corn oil, or a waste stream	Grid electricity, natural gas, and/or renewables	ORRNWD300T	50
	Any feedstock derived from plant oils except for Palm-derived oils	Grid electricity, natural gas, and/or renewables	ORRNWD301T	70
	Distiller's corn oil	Grid electricity, natural gas, and/or renewables	ORRNWD303T	60
	Any feedstock	Any	ORRNWD302T	For 2025: 100.74 For 2026 and beyond: 104.92

Table 9
Oregon Temporary Fuel Pathway Codes for Fuels with Indeterminate CIs

Fuel	Feedstock	Process Energy	FPC	CI (gCO_2e/MJ)
Alternative Jet Fuel	Fats/Oils/Grease residues	Grid electricity, natural gas, and/or renewables	ORAJF350T	55
	Feedstock derived from plant oils (excluding palm oil and palm derivatives, as the sole feedstock or blended with any other feedstock)	Grid electricity, natural gas, and/or renewables	ORAJF351T	75
	Distiller's corn oil	Grid electricity, natural gas, and/or renewables	ORAJF352T	65
	Any other feedstock	Grid electricity, natural gas, and/or renewables	ORAJF353T	Until 2027: 104.92, In 2028 and beyond: 88.21
Renewable Naphtha and Renewable Gasoline Blendstock	Fats/Oils/Grease residues	Grid electricity, natural gas, and/or renewables	ORRNN370T	50
	Feedstock derived from plant oils (excluding palm oil and palm derivatives, as the sole feedstock or blended with any other feedstock)	Grid electricity, natural gas, and/or renewables	ORRNN371T	70
	Distiller's Corn Oil	Grid electricity, natural gas, and/or renewables	ORRNN372T	60
	Any Other Feedstock	Grid electricity, natural gas, and/or renewables	ORRNN373T	For 2025: 100.14, for 2026 and beyond: 98.12
Biomethane CNG	Landfill or Municipal Wastewater sludge	Grid electricity, natural gas, and/or renewables	ORCNG500T	65
	Food Waste, Green Waste, or	Grid electricity, natural gas, and/or parasitic load	ORCNG501T	45

Table 9
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Fuel	Feedstock	Process Energy	FPC	CI (gCO_2e/MJ)
	Other Organic Waste			
Biomethane LNG	Landfill or Municipal Wastewater sludge	Grid electricity, natural gas, and/or renewables	ORLNG501T	80
	Municipal Wastewater sludge, Food Waste, Green Waste, or Other Organic Waste	Grid electricity, natural gas, and/or parasitic load	ORLNG502T	60
Biomethane L-CNG	Landfill or Digester Gas	Grid electricity, natural gas, and/or renewables	ORLCNG502T	90
	Municipal Wastewater sludge, Food Waste, Green Waste, or Other Organic Waste	Grid electricity, natural gas, and/or parasitic load	ORLCNG503T	65
Biomethane CNG, LNG, L-CNG	Dairy and Swine Manure	Grid electricity, natural gas, and/or parasitic load	ORLCNG504T	-150
Fossil Liquified Natural Gas	Fossil Natural Gas	Liquefaction of fossil natural gas using grid electricity and natural gas	ORLNG505T	95
Fossil L-CNG	Fossil Natural Gas	Liquefaction of fossil natural gas that is then decompressed to compressed natural gas	ORLCNG505T	100
Hydrogen (compressed or liquified)	Natural gas	Steam methane reformation using grid electricity/solar and wind electricity, and natural gas with gaseous hydrogen transport distance of less than 500 miles or liquid	ORHYD700T	195

Table 9
Oregon Temporary Fuel Pathway Codes for Fuels with Indeterminate CIs

Fuel	Feedstock	Process Energy	FPC	CI (gCO_2e/MJ)
		hydrogen transport distance of less than 2,000 miles		
	Biomethane from dairy and swine manure	Steam methane reformation using grid electricity/solar and wind electricity and biomethane with gaseous hydrogen transport distance of less than 500 miles or liquid hydrogen transport distance of less than 2,000 miles	ORHYD701T	-40
	Biomethane other than from dairy and swine manure	Steam methane reformation using grid electricity/solar and wind electricity and biomethane with gaseous hydrogen transport distance of less than 500 miles or liquid hydrogen transport distance of less than 2,000 miles	ORHYD702T	175
	Electrolysis of water using zero or negative CI electricity	Electrolysis with gaseous hydrogen transport distance of less than 500 miles or liquid hydrogen transport distance of less than 2,000 miles	ORHYD703T	55
Renewable LPG	Fats, Oils, and Grease residues	Grid electricity, natural gas, and/or renewables	ORRNWP400T	50
	Distiller's corn oil	Grid electricity, natural gas, and/or renewables	ORRNWP402T	60

Table 9
Oregon Temporary Fuel Pathway Codes for Fuels with Indeterminate CIs

Fuel	Feedstock	Process Energy	FPC	CI (gCO_2e/MJ)
	Any feedstock derived from plant oils (excluding palm and palm derivatives)	Grid electricity, natural gas, and/or renewables	ORRNWP401T	70
Electricity	Coal, Natural Gas, Hydroelectric Dams, Windmills, etc.	Oregon average electricity mix	ORELEC600T	150.00
Any Gasoline Substitute Feedstock-Fuel Combination Not Included Above	Any	Any	ORSG800T	For 2025: 100.14, for 2026 and beyond: 98.12
Any Diesel Substitute Feedstock-Fuel Combination Not Included Above	Any	Any	ORSD801T	For 2025: 100.74 For 2026 and beyond: 104.92

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Table 10 Oregon Summary of Indirect Land-Use Change Values for Crop-Based Biofuels	
Feedstock	**ILUC Value (gCO_2e/MJ)**
U.S. Corn Ethanol	7.60
U.S. Sorghum Ethanol	19.40
Brazilian Sugarcane Ethanol	11.80
U.S. Soybean Biodiesel or Renewable Diesel	29.10
North American Canola Biodiesel or Renewable Diesel	14.50
Indonesian and Malaysian Palm Biodiesel or Renewable Diesel	71.40